

COMP3308 / COMP3608 Introduction to Artificial Intelligence

Practice Sample Paper B

Regular-course oriented, paper-style practice. Not an official exam paper.

Time: 2 hours writing time + 10 minutes reading time.

Permitted materials: one double-sided A4 sheet of your own notes; handheld non-programmable calculator.

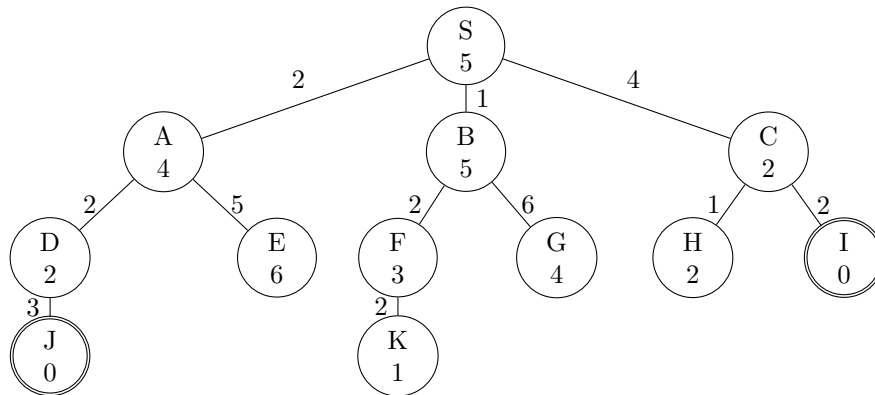
Instructions: Answer all questions in the spaces provided. Use black or blue pen in a real exam. Unless stated otherwise, ties are broken alphabetically or by lower row number. For multiple-choice questions, select all options requested by the question.

Q	Mark
1 /5	
2 /4	
3 /5	
4 /10	
5 /5	
6 /5	
7 /7	
8 /5	
9 /10	
10 /4	

Total **/60**

Question 1. Search algorithms

[5 marks] The graph below is a search tree. Edge labels are step costs. The heuristic value $h(n)$ is shown inside each node. The start node is S . The goal nodes are double-circled: I and J . Stop when a goal node is selected for expansion. In case of ties, expand alphabetically.



Write the node expansion order for each algorithm.

(a) Breadth-first search:

(b) Depth-first search:

(c) Uniform-cost search:

(d) Greedy best-first search:

(e) A* search:

Question 2. Short answers

[4 marks] Answer briefly and precisely.

(a) What is the difference between an admissible heuristic and a consistent heuristic?

(b) Why can a dominant admissible heuristic reduce the number of nodes expanded by A*?

- (c) Give one reason why hill climbing can fail even when a solution exists.

- (d) Alpha-beta pruning may avoid evaluating some leaves. Why does it still return the same minimax decision as full minimax when implemented correctly?

Question 3. Multiple-choice and multiple-select concepts

[5 marks] For each item, circle the correct answer. Some items may require more than one option.

- (a) In uniform-cost search, a goal should be accepted when it is:
- A. first generated
 - B. selected for expansion from the frontier
 - C. assigned the smallest heuristic value
 - D. reached by a depth-first branch
- (b) Select all true statements about Naive Bayes:
- A. It assumes features are conditionally independent given the class.
 - B. It assumes features are unconditionally independent.
 - C. Posterior scores may be compared without normalising if the same evidence is used.
 - D. It cannot be used with numeric attributes.
- (c) K-nearest neighbour is especially sensitive to:
- A. feature scaling
 - B. the class prior only
 - C. the order in which decision-tree attributes are chosen
 - D. alpha and beta bounds
- (d) A single perceptron with a step activation can represent:
- A. any Boolean function
 - B. only linearly separable decision boundaries
 - C. only constant functions
 - D. only probabilistic classifiers
- (e) Which statement about entropy is true?
- A. It is maximised by a uniform distribution over the outcomes.
 - B. It is always larger for a more predictable distribution.
 - C. It is the same as accuracy.
 - D. It is used only for neural networks.

Question 4. Classification, Naive Bayes, KNN and information gain

[10 marks] A fruit classifier is trained from the following ten examples. The class is *good* or *bad*.

n	colour	shape	skin	size	class
1	red	round	smooth	small	good
2	red	round	rough	small	good
3	yellow	long	smooth	large	good
4	yellow	round	smooth	small	good
5	green	round	rough	large	bad
6	green	long	rough	large	bad
7	red	long	rough	large	bad
8	green	round	smooth	small	bad
9	yellow	long	rough	small	good
10	green	long	smooth	large	bad

Classify the new example

$$X = (\text{red}, \text{round}, \text{smooth}, \text{small}).$$

- (a) Use Naive Bayes without Laplace smoothing. Show the class priors, likelihoods, unnormalised scores, and prediction. [4 marks]
- (b) Use 3-nearest neighbour with Hamming distance. Ties in distance are broken by lower row number. Show the three neighbours and the prediction. [2 marks]
- (c) For building a decision tree, calculate the information gain for *colour* and *skin*. Which attribute would be selected among these two? [3 marks]
- (d) Suppose this task is represented for one single perceptron using one-hot encoding for all four categorical attributes. How many input units are needed, excluding the bias input? Briefly explain. [1 marks]

Question 5. Evaluation metrics

[5 marks] A binary classifier for detecting fraudulent transactions returns the following confusion matrix. Treat *fraud* as the positive class.

		Predicted	
		fraud	legitimate
Actual	fraud	32	18
	legitimate	8	42

(a) Compute accuracy. **[1 marks]**

(b) Compute precision for the positive class. **[1 marks]**

(c) Compute recall for the positive class. **[1 marks]**

(d) Compute F_1 . **[1 marks]**

(e) If missing a real fraud case is much worse than wrongly flagging a legitimate case, which metric should receive special attention, and why? **[1 marks]**

Question 6. Perceptron learning

[5 marks] A perceptron has bias input $x_0 = 1$, inputs $(x_1, x_2, x_3) = (1, 0, 1)$, initial weights

$$(w_0, w_1, w_2, w_3) = (-0.1, 0.4, -0.6, 0.2),$$

learning rate $\eta = 0.5$, and step output $y = 1$ if $net \geq 0$, otherwise $y = 0$. The target is $t = 0$.

(a) Calculate the net input. **[1 marks]**

(b) Calculate the perceptron output. **[1 marks]**

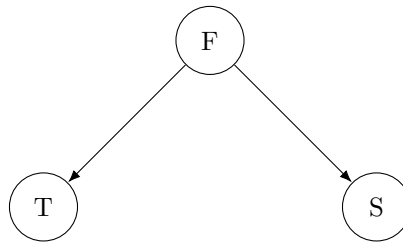
(c) Apply one perceptron learning update $w_i \leftarrow w_i + \eta(t - y)x_i$. Show the new weights. **[2 marks]**

(d) State the main limitation of a single perceptron.

[1 marks]

Question 7. Bayesian networks

[7 marks] Consider the following Bayesian network. F means the patient has flu, T means the test is positive, and S means the patient has symptoms.



The probabilities are:

$$P(F) = 0.1, \quad P(T=+ \mid F) = 0.9, \quad P(T=+ \mid \neg F) = 0.2,$$

$$P(S=true \mid F) = 0.8, \quad P(S=true \mid \neg F) = 0.1.$$

(a) Write the factorisation of the full joint distribution $P(F, T, S)$.

[1 marks]

(b) Compute $P(F \mid T=+, S=true)$. Show numerator, denominator, and final value.

[4 marks]

(c) Are T and S conditionally independent given F ? Are they necessarily marginally independent? Briefly explain.

[2 marks]

Question 8. k-means clustering

[5 marks] Use k-means to cluster the following one-dimensional examples into two clusters:

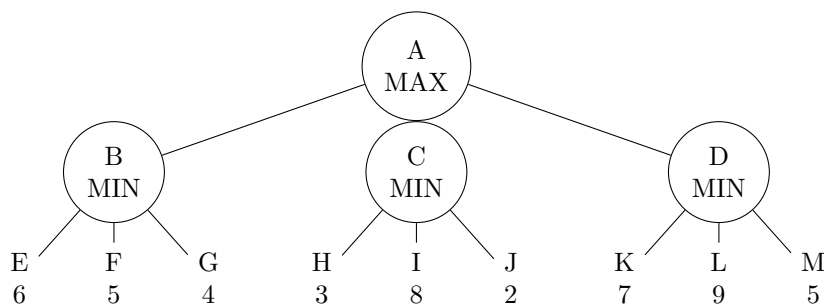
1, 2, 4, 8, 9, 10, 16, 18.

The initial centroids are $c_1 = 2$ and $c_2 = 16$. Assign points to the nearest centroid; break exact distance ties by assigning to the smaller-indexed cluster. Recompute centroids after each epoch. Stop when cluster membership does not change.

Show the cluster membership and centroids for each epoch, then state the final clusters and the number of epochs needed for convergence.

Question 9. Minimax and alpha-beta pruning

[10 marks] In the game tree below, MAX moves at the root A , MIN moves at B, C, D , and the leaf utilities are shown for MAX. Children are visited left-to-right for alpha-beta pruning.



- (a) Compute the backed-up minimax value of each internal node B, C, D, A . **[3 marks]**

- (b) Which move should MAX choose from A ? **[1 marks]**

- (c) Using alpha-beta pruning left-to-right, list the leaf branches that are pruned. You may describe them by leaf labels. **[3 marks]**

- (d) Briefly explain why the pruned leaves cannot change the final decision. **[1 marks]**

- (e) If the children of A were visited in the order D, C, B instead, would alpha-beta generally prune more, less, or the same amount on this tree? Give a brief reason. **[2 marks]**

Question 10. Ensembles, SVM and clustering: select exactly four true statements

[4 marks] Circle exactly four true statements. Each correctly selected true statement is worth 1 mark; selecting more than four options may receive no credit for this question.

- A. Bagging reduces variance by training base learners on bootstrap samples.
- B. Boosting gives larger weights to examples that were previously classified correctly.
- C. Random forests combine bagging with random feature subsets.
- D. In a linear SVM, classifying a new example can involve dot products between the new example and support vectors.
- E. Complete-link cluster distance is the minimum pairwise distance between two clusters.
- F. k-means is guaranteed to find the global optimum for any initial centroids.
- G. Convolutional neural networks use shared filters across spatial locations.
- H. Gain ratio helps reduce information gain's bias toward attributes with many distinct values.